

KOLLÁR SEMI-STABLE GENUS 1 DEGREE 3 MODELS

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ABSTRACT. In this paper we will prove that Kollár's genus 1 equation of degree 3 semi-stable elliptic's over DVR is equivalent to only having canonical singularities.

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1. INTRODUCTION

Notation. In this section, R denotes a discrete valuation ring with quotient field K . Let $t \in R$ be a uniformizer element and let $k = R/(t)$ be the algebraically closed residue field. For a polynomial $F \in R[y_0, \dots, y_n]$, the residue polynomial F_k . Lastly, $F(K)$ is generically smooth.

A genus 1 curve of degree 3 is a tertiary cubic of given form,

$$F(x, y, z) = ax^3 + by^3 + cz^3 + a_2x^2y + a_3x^2z + b_1xy^2 + b_3y^2z + c_1xz^2 + c_2yz^2 + mxyz = 0.$$

Main objective of this investigation to demonstrate an example of a genus 1 model where our notion semi-stability complements arithmetic notion of being minimal, i.e valuation of the discriminant is minimal and birationally producing 'canonical' model which correspond to a simplest singular structure.

The stability condition defined by stability condition was defined by Kollár [7] was an attempt at synthesising several producing 'simple' integral polynomials with integral coefficients and related to GIT. Finding equivalent, such that that the invariant are minimal among all equivalent models, in our particular case discriminant a smooth we will be look g at the discriminant. Second part, of this definition [2] is studied families of Del Pezzo surfaces over smooth curves. His aim was to use birational transformations to obtain a family whose total space has very mild singularities.

One discriminant is a very useful invariant of minimal up to K equivalence for genus 1 models. Sadek [12] has studied in depth geometry of these minimal models up to degree 4 and his paper provides a frame work defined frame work, Kollár stability approximates this notion well. Moreover, in the case $\mathbb{Z}$, if complex smooth then these two notions are equivalent [7, Thm 1.7]. Semi-stability notion extended to weighted hypersurfaces and complete intersection in [1].

Definition 1.1. Let S be a ring. A *weight system* (x, w) is a pair on $R[y_0, \dots, y_n]$ is a choice of coordinates

$$(x_0, \dots, x_n)^t = M(y_0, \dots, y_n)^t,$$

where $M \in \text{GL}(n+1, R)$. And $w \in \mathbb{Z}_{\geq 0}^{n+1}$; w is called the weight vector of the weight system.

We also fix a prime number p and write $v_t(F)$ for the minimum of the p -adic valuations of the coefficients of F .

Definition 1.2 (Unstable Polynomial). A homogeneous polynomial $F \in R[y_0, \dots, y_n]$ is *unstable at t* for a weight system $(\mathbf{x}, \mathbf{w})$ with $w = [w_0, \dots, w_n]$ if

$$v_t(F(t^{w_0}x_0, t^{w_1}x_1, \dots, t^{w_n}x_n)) > \frac{d}{n+1} (w_0 + w_1 + \dots + w_n),$$

where v_t denotes the t -adic valuation. **Semi-stable** if inequality does not hold for any weight system.

Algorithms for producing simple models have two approaches. One is in terms reduction, which reduces the valuation reduces, i.e 'smaller coefficients', while staying in the same isomorphism class over R . The other is minimisation, which reduces the valuation of associated invariants of the polynomial. Minimization and reduction have been studied for 2-, 3-, 4- and 5-coverings of elliptic curves in [3] and [5].

The following procedure developed by Kollár, produces semi-stable models but not necessarily minimal.

Take our integral model F_j and assume unstable w.r.t weight system $(\mathbf{x}, \mathbf{w})$ an integral weight system unstable on F_j , then set

$$F_{j+1} := t^{-s} F_j(t^{w_0}x_0, \dots, t^{w_n}x_n), \quad s := v_t(F_j(t^{\mathbf{w}}\mathbf{x})).$$

For any invariant Δ_j , of F_j we see that $v_t(\Delta_{j+1}) \leq v_t(\Delta_j)$, thus ensuring $F(K)$ is generically smooth directs us to minimal models.

For surfaces, the mildest singularities are the *canonical singularities*, also known as *Du Val singularities*, *rational double points*, *Kleinian singularities*, or *simple singularities*. These are hypersurface singularities classified by the Dynkin types A_n , D_n , E_6 , E_7 , E_8 , and admit explicit local defining equations [11]. A surface whose singularities are all canonical is called a *canonical model*.

Theorem 1.3. *Genus 1 degree 3 models are semi-stable iff any singularity is Du Val and stable iff smooth*

2. DISCUSSION ON INVARIANT THEORY

Let K be an infinite field, and let $f \in K[y_0, \dots, y_n]$ be a homogeneous polynomial of degree d . Furthermore, let homogeneous polynomial I with integral coefficients in the coefficients in K of the form F such that $I(f(My)) = I(f(x))$ for all $M \in SL_{n+1}(K)$.

Theorem 2.1. [9, pp. 49 and 79] *Semistability in Geometric Invariant Theory There exists an SL -invariant polynomial I such that $I(f) \neq 0$ if and only if f is semistable.*

It follows that in the Kollár stability setting. To semi-stability is 'equivalent' to minimising an invariant I over DVR with integral coefficients. Thus, if a our fibers are GIT semi-stable, our exists an invariant $F \in R[y]$ that is $v_t(I) = 0$, since then moding by our prime p is exactly the existence of a non-zero invariant $I_k(I \bmod k)$.

Proposition 2.2. [7, proposition 3.5] *Assume that $f_k \in k[y_0, \dots, y_n]$ is GIT semistable (resp. properly stable) over $\bar{k}$. Then F_R is semistable (resp. properly stable) at p .*

The following details semi-stability of plane cubics.

Theorem 2.3. *A plane cubic curve C defined by $f \in k[x_0, x_1, x_2]_3$ is semi-stable if and only if it Hessian rank 2. A plane cubic curve C is stable if and only if it is smooth.*

Proof. For wieghts we can assume wlog that weight of the form $\mathbf{w} = (w_0, w_1, w_2)$ and $f = \sum_{i=0}^3 \sum_{j=0}^{3-i} a_{ij} x_0^{3-i-j} x_1^i x_2^j$, thus to be unstable. Take meaning unsstable if $v_t(F) = \min\{(2-i-j)r_0 + (i-1)r_1 + (j-1)r_2\} > 0$. We can assume that's is non trivial $w_0 \leq w_1 \leq w_2$. Therefore we can immediately conclude that $a_{0,0} = a_{1,0} = a_{0,1}$ and singular at the point $[0 : 0 : 1]$.

If $a_{1,1} \neq 0$, then rank will be 2, but this will contradict our assumption because $v_t(F) = 0$.

It follows that the local hessian matrix at the point $[1 : 0 : 0]$.

$$H_p(F) = \begin{pmatrix} \frac{\partial^2 F}{\partial x_1^2}(0,0) & \frac{\partial^2 F}{\partial x_1 \partial x_2}(0,0) \\ \frac{\partial^2 F}{\partial x_2 \partial x_1}(0,0) & \frac{\partial^2 F}{\partial x_2^2}(0,0) \end{pmatrix} = \begin{pmatrix} 2a_{10} & a_{11} \\ a_{11} & 2a_{01} \end{pmatrix},$$

however, due to our inequality that $a_{2,0} = a_{1,1} = 0$, thus of rank at most 1.

Conversely, we assume that rank is at 1, then split into two cases. $a_{1,0} = a_{1,1} = 0$, then unstable with respect to weight $(0, 2, 3)$.

Lastly, if we there singular point at $[1 : 0 : 0]$, so $a_{0,0} = a_{0,1} = 1, 0 = 0$ then testing weight $(0, 1, 1)$ will at imply not stable. On the other hand again, if not stable then $\min\{(2-i-j)r_0 + (i-1)r_1 + (j-1)r_2\} \geq 0$. Now for any no trivial weight system, we see that $r_2 > r_0$ and $r_2 \geq r_1$. Hence, explicitly we see $a_{0,0} = 0$, due to $2r_0 - (r_1 + r_2) < 0$. Similar reasoning to prove that $a_{0,1} = a_{1,0} = 0$.

□

Please see [6, Lemma 6.17] for an equivalent statement and a more detailed discussion of the various cases, including explicit examples.

3. SEMI-STABLE MODELS OF ELLIPTIC CURVES ARE NORMAL

As stated our notion semi-stability approximates a model being minimal, and in this chapter we show how it interpolates normality aswell. Methods and proof adapted from [12, Section 3]. We assume models will be $C = (F = 0)$, where $F \in R[x_0, x_1, x_2]_3$ and C_k central fibre.

To check the semi-stability of a family of cubic surfaces over a one-dimensional regular scheme, it is sufficient to consider only the five weight sequences

$$(0, 0, 1), \quad (0, 1, 1), \quad (0, 1, 1), \quad (0, 1, 2), \quad (0, 2, 3),$$

as shown by Elsenhans and Stoll [4, Table 1]. Furthermore, this paper shows that for a degree d hypersurface in any dimension n and degree d , the existence of a finite weight set to determine semi-stability is evident and gives an explicit bound for weight entries.

It follows for $F \in R[x_0, x_1, x_2]_3$, in $\mathbb{P}_R^2$, is unstable if

$$v_t(F'(t^{w_0}x_0, t^{w_1}x_1, t^{w_2}x_2)) > w_0 + w_1 + w_2$$

, with respect to weight system (σ, w) .

Lemma 3.1. *semi-stable cubic hypersurface $X \subset \mathbb{P}_R^n$ do not contain triple points*

Proof. We can assume that our triple point is at $[0 : 0 : \dots : 0 : 1]$, thus unstable with weight vector $(1, 1, \dots, 1, 0)$, as $v_t(F) = 3 > \frac{3}{n+1}n \geq 2$. □

Another form of writing our genus 1 degree 3 model C is,

$$\phi : bx_0^3 + f_1(x_0, x_1)x_2^2 + f_2(x_0, x_1)x_2 + f_3(x_0, x_1) = 0,$$

with $f_1(x_0, x_1) = b_1x_0 + b_3x_1$, $f_2(x_0, x_1) = a_2x_0^2 + mx_0x_1 + c_2x_1^2$, and $f_3(x_0, x_1) = ax_0^3 + a_3x_0^2x_1 + c_1x_0x_1^2 + cx_1^3$. If C_k contains an irreducible component of multiplicity m , $m \geq 2$, then we can assume without loss of generality that the defining is $y = 0$.

Proposition 3.2. [12, Proposition 3.3]

(i) *If C_k is reduced, then C is normal.*

(ii) *If $n = 3$, $\phi : F(x_0, x_1, x_2) = 0$ is given as in equation (4), and C_k contains a multiplicity- $m \geq 2$ component $\Gamma : \{x_2 = 0\}$, then C is normal if and only if $v_t(f_3) = 1$.*

Theorem 3.3. *If C semi-stable model, then it is normal.*

Proof. C_k is reduced and then C is normal.

In the other case suppose that C_k is not reduced, i.e contains a irreducible component of multiplicity greater than 1 and C is not normal. It follows by the previous proposition, since C is not normal, we have $v_t(f_2) \geq 1$ and $v_t(f_3) \geq 2$. Thus, unstable with respect to weight $(0, 0, 1)$. □

4. SEMI-STABLE ELLIPTIC MODELS ARE CANONICAL

4.1. Introduction to Newton Polytopes. We fix the following notions and notation. Let $f \in k[[x_1, \dots, x_n]]$ be a formal power series which is in-decomposable into non-invertible factors and has trivial 1-jet.

The *Newton polyhedron* of a power series

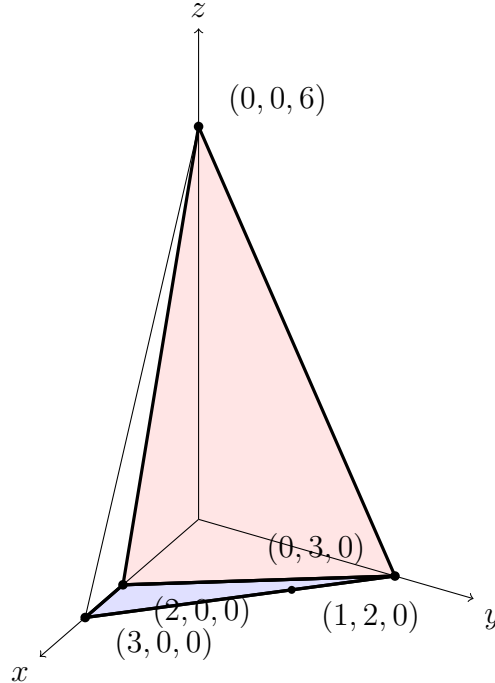
$$f = \sum_m a_m x^m$$

is defined as the convex hull in $\mathbb{R}^n = \mathbb{Z}^n \otimes_{\mathbb{Z}} \mathbb{R}$ of the set of exponents $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$ such that $a_m \neq 0$, where $x^m = x_1^{m_1} \cdots x_n^{m_n}$. For the power series f , we denote by $\Delta(f)$ the *Newton diagram*, which is the union of all finite faces of the Newton polyhedron facing the origin.

Definition 4.1. A point $P \in \mathbb{R}^n$ is said to lie *above* a set $K \subset \mathbb{R}^n$ if for every point $Q \in K$, the difference $P - Q$ has strictly positive coordinates.

The next few examples will be the newton polytopes of unstable equations localised at the point $[0 : 0 : 1]$

Example 4.2. Unstable with respect to weight $(3, 2, 0)$ is given by the polynomial $x^2 + x^3 + y^3 + xy^2 + t^6$.



$(1, 1, 1)$ lies red region of lower convex hull. Described by the plane $3x + 2y + z = 6$, thus cannot be canonical.

Example 4.3. Unstable with respect to weight $(2, 1, 0)$, local equation is given by $x^2 + xy^2 + y^2t^2 + t^4 = 0$.

$(1, 1, 1)$ lies in the blue region, lower convex hull. is described by plane $x + y = 2$, thus cannot be canonical.

Example 4.4. Unstable with respect to weight $(1, 1, 0)$, local equation is given by $bx^3 + y^3 + t^3 + t^2xy + t^4 = 0$.

$(1, 1, 1)$ lies in blue region of lower convex hull described by equation $x + y + z = 3$. Singularity is not canonical.

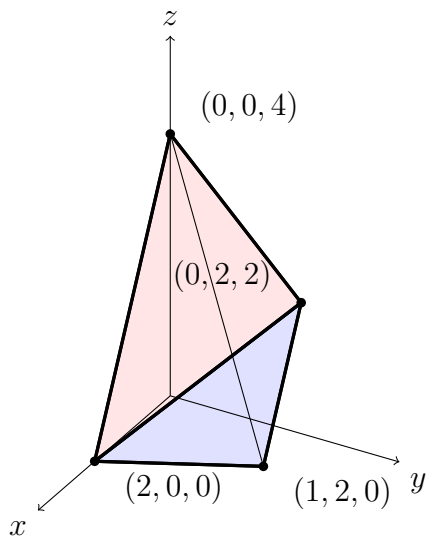


FIGURE 1. Newton polytope $x^2 + xy^2 + y^2t^2 + t^4$

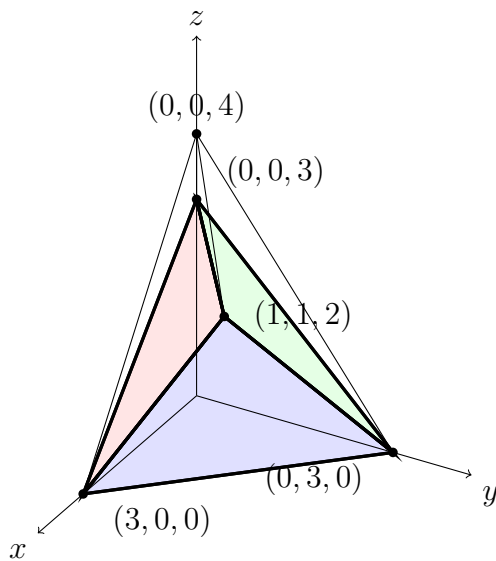


FIGURE 2. Newton Polytope: $x^3 + y^3 + t^3 + t^2xy + t^4$

From the following theorem we see previous examples are not canonical. Also, and will help prove [Thm 1.3].

Theorem 4.5. [8, Theorem 1.2] *A necessary condition for a hypersurface $\{f = 0\} \subset k^n$, $f = \sum_m a_m x^m$, to have a canonical singularity at zero is that the point $(1, \dots, 1)$ lies above the Newton diagram $\Delta(f)$ of the function f .*

4.2. Proof of theorem 1.3. In this last subsection, we the equivalenc of semi-stable and canonical models..

Proposition 4.6. [7, proposition 3.5] and [10, lemma 6.2]

Let $\hat{R}$ denote the completion of R at (p) . Then:

- (1) $f_R \in R[y]$ is properly stable (semi-stable, unstable) at $p \in R$ if and only if $f_R \in \hat{R}[y]$ is properly stable (semi-stable, unstable) at $p \in \hat{R}$.
- (2) $\text{char}(k) = \text{char}(R)$ then $\hat{R} \simeq k[[t]]$.

If your equations are not generic enough, after what is a sufficient condition that makes a unstable equation such that singular point in non-degenerate.

Proposition 4.7. [1, lemma 5.3] Let $f \in k[[x, y, z]]$ define a hypersurface $X = \text{Spec } k[[x, y, z]]/(f)$ with an isolated singularity of multiplicity 2 at origin P .

Then the following hold:

- (1) If the tangent cone of f is reduced, then f is formally equivalent to $xy + z^{n+1} = 0$, that is, P is a singularity of type A_n for some $n \geq 1$.
- (2) If $f = x^2 + g(y, z)$ and the tangent cone of $g(y, z)$ is a cubic with at least two distinct linear factors, then f is formally equivalent to $x^2 + z(y^2 + z^{n-2}) = 0$, that is, P is a singularity of type D_n for some $n \geq 4$.
- (3) If $f = x^2 + y^3 + h(y, z)$, $\deg h \geq 4$ and $\text{val}(h(t^2y, tz)) \leq 5$, equivalently, if $h(y, z)$ contains a term z^4 , yz^3 , or z^5 , then P is a singularity of type E_6 , E_7 , or E_8 .

We now have all the necessary ingredients to prove the main theorem of this paper.

Theorem 4.8. Genus 1 degree 3 models are semi-stable iff any singularity is Du Val and stable iff smooth

Proof. In the case of an isolated real singularity, consider a monomial $x^a y^b t^c$. Assuming the singular point is fixed at $[0 : 0 : 1]$, for the point to be unstable, we consider a weight system $(k_1, k_2, 0)$ where $k_i \geq 1$. The set of unstable coordinates is given by:

$$S_{k_1, k_2} = \{(a, b, c) \mid k_1 a + k_2 b + c > k_1 + k_2\}$$

The central question is whether the point $(1, 1, 1)$ lies in the interior of this set or on the boundary. It is clear that $(1, 1, 1)$ lies, at best, on the boundary (where $k_1(1) + k_2(1) = k_1 + k_2$). Consequently, the singularity is not canonical because it fails the strictly destabilizing condition.

For the converse, assume singularity at $[0 : 0 : 1]$, thus local equation is given by $f \in k[[x, y, t]]$, degree f is at most 3 in x, y . Also, semi-stability implies double point by lemma 3.1. Hence, will go through case Du Val and show unstable.

- **Case 1:** Suppose tangent cone is notreduced (i.e., formally equivalent to xy or xt), i.e rank 2, proceed next two cases.
- **Case 2:** We f is of the form $x^2 + g(y, t)$ where the tangent cone of g is a cubic with one linear factor. Therefore (e.g., t^3 or x_1^3). . In $k[[t]][x_0, x_1]_{\leq 3}$ and it follows that $x^2 + (t^3 \text{ or } y^3) + \hat{g}(y, t)$ where $\hat{g}$ is degree 4.
 - If we have $x^2 + y^3 + \hat{g}(y, t)$. If $t^4, yt^3, t^5 \in \hat{g}$ then Du Val by case 3. Otherwise, unstable w.r.t $(3, 2, 0)$.

- If we have $x^2 + t^3 + \hat{g}(y, t)$ changes, as if we have $y^3 t \in \hat{g}$ then Du Val by case 2. Otherwise $t^2 \mid \hat{g}$, hence unstable w.r.t $(1, 0, 0)$.
- **Case 3:** Again by similar reasoning as previous case, where $f = x^2 + y^3 + h(y, t)$ or $x^2 + t^3 + h(y, t)$ where $\deg h < 4$ or $v_s(h(s^2 y, st))$.
 - If degree $\deg h < 3$, this has already been shown dealt with by previous two cases.
 - If we have $x^2 + y^3 + h(y, t)$ where $v_s(s^2 y, st) > 5$, then unstable w.r.t $(3, 2, 0)$. Or, $v_s(h(s^2 t, sy)) > 5$ if we have $t^3, t^4, t^5, yt^3 \in h$ is Du Val. Otherwise, unstable with respect to $(3, 2, 0)$.
 - If we have $x^2 + t^3 + h(y, t)$, where $v_s(h(s^2 y, st)) > 5$. If $y^3, y^3 t \in h$ then it will be Du Val. Otherwise, unstable w.r.t $(1, 0, 0)$. Or $v_s(h(s^2 t, sy)) > 5$, then $t^2 \mid h$, hence unstable w.r.t $(1, 0, 0)$.

We can assume singular point is at $[0 : 0 : 1]$. Thus at best semi-stable w.r.t $(1, 1, 0)$, as $v_t(F) \geq 2$. On the other hand, for a non-trivial weight set of the form $(a, b, 0)$ where a or $b \geq 1$, and not stable then $v_t F(ax, bx_2, x_0) \geq a + b$. Implies $F \in \langle x_0, x_1, t \rangle^2$, hence singular. $\square$

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